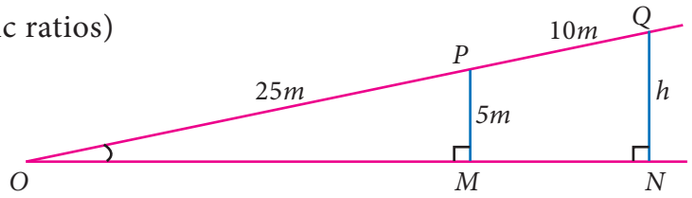
10. A boy standing at a point O finds his kite flying at a point P with distance $OP=25m$. It is at a height of $5m$ from the ground. When the thread is extended by $10m$ from P, it reaches a point Q. What will be the height QN of the kite from the ground? (use trigonometric ratios)



6.2 Trigonometric Ratios of Some Special Angles

The values of trigonometric ratios of certain angles can be obtained geometrically. Two special triangles come to help here.

6.2.1 Trigonometric ratios of 45°

Consider a triangle ABC with angles 45°, 45° and 90° as shown in the figure 6.9.

It is the shape of half a square, cut along the square's diagonal. Note that it is also an isosceles triangle (both legs have the same length, a units).

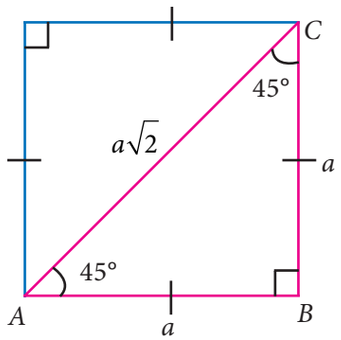


Fig. 6.9

Use Pythagoras theorem to check if the diagonal is of length $a\sqrt{2}$.

Now, from the right-angled triangle ABC,

$$\sin 45^o = \frac{opposite\ side}{hypotenuse} = \frac{BC}{AC} = \frac{a}{a\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$\cos 45^o = \frac{adjacent\ side}{hypotenuse} = \frac{AB}{AC} = \frac{a}{a\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$\tan 45^o = \frac{opposite\ side}{adjacent\ side} = \frac{BC}{AB} = \frac{a}{a} = 1$$

The reciprocals of these ratio can be easily foundout to be

$\operatorname{cosec} 45° = \sqrt{2}$;

$\sec 45° = \sqrt{2}$ and

$\cot 45° = 1$

6.2.2 Trigonometric Ratios of 30° and 60°

Consider an equilateral triangle PQR of side length 2 units.

Draw a bisector of $\angle P$. Let it meet QR at M.

$PQ = QR = RP = 2$ units.

$QM = MR = 1$ unit (Why?)

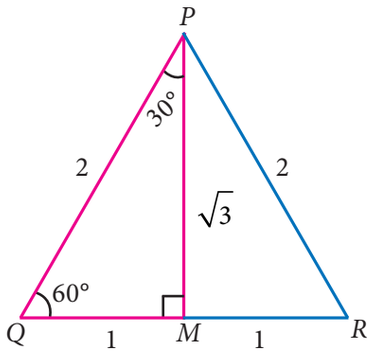


Fig. 6.10

Knowing PQ and QM, we can find PM, using Pythagoras theorem,

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